JAVA OBJECT-ORIENTED PROGRAMMING

GENERAL INSTRUCTIONS

For each question, students are required to:

1. Develop an application containing five Java classes, including the Main class.
2. Ensure that each of the four domain classes contains exactly five attributes.
3. Apply encapsulation, constructors, getters and setters.
4. Establish appropriate relationships between the classes.
5. Override the `toString()` method in all four domain classes.
6. Allow the user to enter information during program execution.
7. Register objects in the specified order.
8. Implement appropriate validation for all attributes.
9. Handle invalid input and exceptions without unexpected program termination.
10. Allow the user to correct invalid information.
11. Display all registered objects only after successful completion of all registrations.
12. Independently determine the appropriate implementation approach.

QUESTION 1: LIBRARY MANAGEMENT SYSTEM

Scenario

A university library wants to develop a system to manage books, library members, librarians and borrowing transactions.

Develop a Library Management System using the following five classes.

Class 1: Book

Attribute	Data Type
bookId	int
bookTitle	String
author	String
publicationYear	int
availableCopies	int

Class 2: Member

Attribute	Data Type

memberId	int
memberName	String
memberEmail	String
memberAge	int
membershipType	String

Class 3: Librarian

Attribute	Data Type
librarianId	int
librarianName	String
librarianEmail	String
yearsOfExperience	int
librarySection	String

Class 4: Borrowing

Attribute	Data Type
borrowingId	int
book	Book
member	Member
librarian	Librarian
borrowingDays	int

Class 5: Main

The application must register information in the following order:

Book → Member → Librarian → Borrowing.

All registered information must be validated and displayed only after completing the entire registration process.

QUESTION 2: HOSPITAL MANAGEMENT SYSTEM

Scenario

A hospital wants to develop an application to manage doctors, patients, hospital departments and medical appointments.

Develop the application using the following five classes.

Class 1: Department

Attribute	Data Type
departmentId	int
departmentName	String
departmentCode	String
floorNumber	int
departmentLocation	String

Class 2: Doctor

Attribute	Data Type
doctorId	int
doctorName	String
specialization	String
phoneNumber	String
department	Department

Class 3: Patient

Attribute	Data Type
patientId	int
patientName	String
patientAge	int
patientPhone	String
patientGender	String

Class 4: Appointment

Attribute	Data Type
appointmentId	int
doctor	Doctor
patient	Patient
appointmentDate	String
appointmentStatus	String

Class 5: Main

Registration order: Department → Doctor → Patient → Appointment.

The application must validate all information, maintain the required relationships and display all registered objects after successful registration.

QUESTION 3: BANK MANAGEMENT SYSTEM

Scenario

A commercial bank wants to develop an application to manage bank branches, customers, accounts and financial transactions.

Develop the application using the following five classes.

Class 1: BankBranch

Attribute	Data Type
branchId	int
branchName	String
branchCode	String
branchLocation	String
branchManager	String

Class 2: Customer

Attribute	Data Type
customerId	int
customerName	String
customerEmail	String
customerPhone	String
customerAge	int

Class 3: BankAccount

Attribute	Data Type
accountNumber	String
accountType	String
balance	double
customer	Customer
branch	BankBranch

Class 4: Transaction

Attribute	Data Type
transactionId	int
account	BankAccount
transactionType	String
transactionAmount	double
transactionDate	String

Class 5: Main

Registration order: BankBranch → Customer → BankAccount → Transaction.

The system must ensure that all banking information is valid and that transactions are associated with registered accounts.

QUESTION 4: EMPLOYEE MANAGEMENT SYSTEM

Scenario

A company wants to develop an application to manage departments, employees, projects and employee assignments.

Develop the application using the following five classes.

Class 1: Department

Attribute	Data Type
departmentId	int
departmentName	String
departmentCode	String
departmentLocation	String
departmentManager	String

Class 2: Employee

Attribute	Data Type
employeeId	int
employeeName	String
employeeEmail	String
salary	double
department	Department

Class 3: Project

Attribute	Data Type
projectId	int
projectName	String
projectBudget	double
projectDuration	int
projectStatus	String

Class 4: Assignment

Attribute	Data Type
assignmentId	int
employee	Employee
project	Project
assignedHours	int
assignmentStatus	String

Class 5: Main

Registration order: Department → Employee → Project → Assignment.

The system must validate all registered information and ensure that project assignments reference valid employees and projects.

QUESTION 5: HOTEL RESERVATION MANAGEMENT SYSTEM

Scenario

A hotel wants to develop an application to manage hotel branches, rooms, guests and reservations.

Develop the application using the following five classes.

Class 1: Hotel

Attribute	Data Type
hotelId	int
hotelName	String
hotelLocation	String

hotelRating	int
hotelPhone	String

Class 2: Room

Attribute	Data Type
roomNumber	int
roomType	String
roomPrice	double
roomCapacity	int
hotel	Hotel

Class 3: Guest

Attribute	Data Type
guestId	int
guestName	String
guestEmail	String
guestPhone	String
guestAge	int

Class 4: Reservation

Attribute	Data Type
reservationId	int
guest	Guest
room	Room
numberOfNights	int
reservationStatus	String

Class 5: Main

Registration order: Hotel → Room → Guest → Reservation.

The application must ensure that all reservations contain valid information and reference registered guests and rooms.

QUESTION 6: VEHICLE RENTAL MANAGEMENT SYSTEM

Scenario

A vehicle rental company wants to develop an application to manage rental branches, vehicles, customers and rental agreements.

Develop the application using the following five classes.

Class 1: RentalBranch

Attribute	Data Type
branchId	int
branchName	String
branchLocation	String
branchPhone	String
branchManager	String

Class 2: Vehicle

Attribute	Data Type
vehicleId	int
plateNumber	String
vehicleBrand	String
dailyRentalPrice	double
branch	RentalBranch

Class 3: Customer

Attribute	Data Type
customerId	int
customerName	String
customerPhone	String
customerEmail	String
drivingLicenseNumber	String

Class 4: Rental

Attribute	Data Type
rentalId	int
vehicle	Vehicle
customer	Customer

rentalDays	int
rentalStatus	String

Class 5: Main

Registration order: RentalBranch → Vehicle → Customer → Rental.

The application must validate all rental information and ensure that rental agreements reference registered vehicles and customers.

QUESTION 7: ONLINE SHOPPING MANAGEMENT SYSTEM

Scenario

An online shopping company wants to develop an application to manage product categories, products, customers and purchase orders.

Develop the application using the following five classes.

Class 1: Category

Attribute	Data Type
categoryId	int
categoryName	String
categoryCode	String
categoryDescription	String
categoryStatus	String

Class 2: Product

Attribute	Data Type
productId	int
productName	String
unitPrice	double
availableQuantity	int
category	Category

Class 3: Customer

Attribute	Data Type

customerId	int
customerName	String
customerEmail	String
customerPhone	String
deliveryAddress	String

Class 4: Order

Attribute	Data Type
orderId	int
product	Product
customer	Customer
orderedQuantity	int
orderStatus	String

Class 5: Main

Registration order: Category → Product → Customer → Order.

The application must ensure that all orders contain valid information and are associated with registered products and customers.

QUESTION 8: SCHOOL MANAGEMENT SYSTEM

Scenario

A secondary school wants to develop an application to manage academic departments, teachers, subjects and teaching assignments.

Develop the application using the following five classes.

Class 1: Department

Attribute	Data Type
departmentId	int
departmentName	String
departmentCode	String
departmentHead	String
officeLocation	String

Class 2: Teacher

Attribute	Data Type
teacherId	int
teacherName	String
teacherEmail	String
qualification	String
department	Department

Class 3: Subject

Attribute	Data Type
subjectId	int
subjectName	String
subjectCode	String
creditHours	int
educationLevel	String

Class 4: TeachingAssignment

Attribute	Data Type
assignmentId	int
teacher	Teacher
subject	Subject
academicYear	int
assignedHours	int

Class 5: Main

Registration order: Department → Teacher → Subject → TeachingAssignment.

The application must validate all academic information and ensure that teaching assignments reference registered teachers and subjects.

QUESTION 9: REAL ESTATE MANAGEMENT SYSTEM

Scenario

A real estate company wants to develop an application to manage property owners, properties, tenants and rental contracts.

Develop the application using the following five classes.

Class 1: PropertyOwner

Attribute	Data Type
ownerId	int
ownerName	String
ownerEmail	String
ownerPhone	String
ownerAddress	String

Class 2: Property

Attribute	Data Type
propertyId	int
propertyLocation	String
propertyType	String
monthlyRent	double
owner	PropertyOwner

Class 3: Tenant

Attribute	Data Type
tenantId	int
tenantName	String
tenantEmail	String
tenantPhone	String
tenantAge	int

Class 4: RentalContract

Attribute	Data Type
contractId	int
property	Property
tenant	Tenant
rentalDuration	int
contractStatus	String

Class 5: Main

Registration order: PropertyOwner → Property → Tenant → RentalContract.

The application must ensure that rental contracts are associated with valid properties and tenants and that all registered information satisfies appropriate validation requirements.

QUESTION 10: UNIVERSITY EXAMINATION MANAGEMENT SYSTEM

Scenario

A university wants to develop an application to manage academic departments, lecturers, students and examination results.

Develop the application using the following five classes.

Class 1: Department

Attribute	Data Type
departmentId	int
departmentName	String
departmentCode	String
departmentHead	String
officeLocation	String

Class 2: Lecturer

Attribute	Data Type
lecturerId	int
lecturerName	String
lecturerEmail	String
specialization	String
department	Department

Class 3: Student

Attribute	Data Type
studentId	int
studentName	String
studentEmail	String
studentAge	int
department	Department

Class 4: ExaminationResult

Attribute	Data Type
resultId	int
student	Student
lecturer	Lecturer
examinationMarks	double
examinationStatus	String

Class 5: Main

Registration order: Department → Lecturer → Student → ExaminationResult.

The application must validate all academic information and ensure that examination results reference registered students and lecturers.